

Relations & Functions – Quick Reference Sheet

Extracted from solved ISC problems (Ch. 1) | ISC Class XII

A. Relations – Definitions & Counting

$$n(R) = 2^{n^2}$$

$$n(R_{\text{reflexive}}) = 2^{n^2 - n}$$

$$\text{Reflexive: } (a, a) \in R \quad \forall a \in A$$

$$\text{Symmetric: } (a, b) \in R \implies (b, a) \in R$$

$$\text{Transitive: } (a, b), (b, c) \in R \implies (a, c) \in R$$

$$\text{Equivalence relation} = \text{reflexive} + \text{symmetric} + \text{transitive}$$

$$[a] = \{x \in A : (x, a) \in R\}$$

Tip – R/S/T Checking Strategy

Test *reflexive* first (every diagonal pair (a, a) must appear). For *symmetric* and *transitive*, don't try to prove they hold everywhere – actively hunt for ONE counter-example pair; a single failure is enough to disprove either property.

B. Domain & Range Rules

$$\text{Even root: } g(x) \geq 0$$

$$\text{Denominator} \neq 0$$

$$\log(g(x)) \text{ defined} \iff g(x) > 0$$

$$a \sin x + b \cos x = r \sin(x + \alpha), \quad r = \sqrt{a^2 + b^2}$$

Tip – Auxiliary Angle Method

For $a \sin x + b \cos x$: put $a = r \cos \alpha$, $b = r \sin \alpha$ where $r = \sqrt{a^2 + b^2}$. Since $-1 \leq \sin(x + \alpha) \leq 1$, the range is exactly $[-r, r]$. (A single term like $5 \sin 3x + 2$ needs no auxiliary angle – just shift the standard $[-1, 1]$ range.)

C. Standard Functions (Domain / Range)

Quick Table – Common Named Functions

Identity: $f(x) = x$	Domain = $\mathbb{R}$, Range = $\mathbb{R}$
Constant: $f(x) = c$	Domain = $\mathbb{R}$, Range = $\{c\}$
Modulus: $f(x) = x $	Domain = $\mathbb{R}$, Range = $[0, \infty)$
Signum: $f(x) = \frac{ x }{x}$ ($x \neq 0$), $f(0) = 0$	Range = $\{-1, 0, 1\}$
Greatest integer: $f(x) = [x]$	Domain = $\mathbb{R}$, Range = $\mathbb{Z}$

D. Types of Functions – Tests & Counting

$$f \text{ one-one} \iff f(x_1) = f(x_2) \implies x_1 = x_2$$

$$f: A \rightarrow B \text{ onto} \iff \forall y \in B, \exists x \in A : f(x) = y$$

$$\text{Functions } A \rightarrow B = n^m \quad (m = |A|, n = |B|)$$

$$\text{One-one functions} = {}^n P_m \quad (m \leq n); \quad 0 \quad (m > n)$$

$$\text{Onto } (n=2): 2^m - 2; \quad (n=3): 3^m - 3(2^m - 1)$$

$$\text{Bijections} = m! \quad (m = n); \quad 0 \quad (m \neq n)$$

Tip – Which Counting Formula?

Fix which set is the domain (size m) and which is the codomain (size n) before substituting – it's easy to swap them by mistake. When $m = n$, a one-one function is automatically onto too, so it simplifies straight to $m!$.

E. Composition of Functions

$$(g \circ f)(x) = g(f(x))$$

$$f, g \text{ one-one} \implies g \circ f \text{ one-one}$$

$$f, g \text{ onto} \implies g \circ f \text{ onto}$$

Tip – Order of Composition

$(g \circ f)(x) = g(f(x))$: work out $f(x)$ first, then plug that ENTIRE result into g . Composition is generally NOT commutative ($g \circ f \neq f \circ g$) – never assume, always compute both sides. If both composites simplify to plain x (i.e. $g \circ f = f \circ g = I$), the

two functions are mutual inverses.

F. Invertible Functions

f invertible $\iff f$ bijective (one-one & onto)

Put $y = f(x)$, solve for $x \implies f^{-1}(y) = x$

$$(f^{-1})^{-1} = f$$

f self-inverse $\iff f^{-1}(x) = f(x)$ (check $f(f(x)) = x$)

$$f^{-1} = \{(b, a) : (a, b) \in f\}$$

Worked Pattern – Restricting Domain/Codomain to Invert

Quadratic on $\mathbb{N}$ (e.g. $f(x) = 4x^2 + 12x + 15$): complete the square first: $4x^2 + 12x + 15 = (2x+3)^2 + 6$. Since $x \in \mathbb{N}$ makes $(2x+3) > 0$, only the *positive* root applies when inverting – this makes f one-one, and restricting the codomain to $S = \text{range}(f)$ makes it onto automatically. Then $f^{-1}(y) = \frac{\sqrt{y-6}-3}{2}$.

Non-monotonic cubic (e.g. $f(x) = 2x^3 - 3x^2 - 12x + 8$): a wavy graph is never one-one on all of $\mathbb{R}$. Find turning points by solving $f'(x) = 0$ – these bound the LARGEST interval containing the required point on which f is monotonic (hence invertible). Then domain of $f^{-1} = \text{range of } f \text{ on that interval}$, and range of $f^{-1} = \text{the restricted interval itself}$.

Quick Recognition Guide

Check R / S / T of a relation

Prove equivalence relation + find class

Domain/range: surd, rational, modulus

Domain: log / trig expression

$a \sin x + b \cos x$ type

Prove one-one

Prove onto / not onto

Counting functions/bijections

Composition of algebraic functions

Prove mutual inverses

Find inverse (algebraic)

Invertibility from graph/diagram

Restrict codomain (quadratic on $\mathbb{N}$)

Restrict domain (non-monotonic function)

Test all 3 directly; hunt a counter-example for S, T

Prove R,S,T for general elements, then test the condition against every element

Domain: root ≥ 0 , denom $\neq 0$; range: simplify fully, track excluded values

Combine log > 0 , trig inequality, denom $\neq 0$ together

Auxiliary angle, range $[-r, r]$, $r = \sqrt{a^2 + b^2}$

Assume $f(x_1) = f(x_2)$, solve to $x_1 = x_2$

Solve $f(x) = y$ for x ; check x lies in the domain

Pick right formula: n^m , ${}_nP_m$, onto shortcut, or $m!$

Substitute the ENTIRE inner function into the outer

Show $g \circ f = f \circ g = I$

Set $y = f(x)$, solve for x

2 arrows/points sharing a target $\Rightarrow$ not one-one; vertical line hits twice $\Rightarrow$ not even a function

Complete the square, take positive root only

Find turning points via $f'(x) = 0$, take largest interval